

Differentiating Colon and Rectum Adenocarcinoma with Microbiome-Based Machine Learning Models

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Abstract

Colon and rectum cancer are often grouped together under colorectal cancer due to their similarity; however, treatment and patient outcomes differ. Therefore, the ability to correctly differentiate between the two is essential. This paper assesses how machine learning can classify colon versus rectum cancer using microbiome data from The Cancer Microbiome Atlas (TCMA). Specifically, I use Logistic Regression, Support Vector Machine, and Multi-Layer Perceptron to predict which project within TCMA the sample is from - either colon or rectum adenocarcinoma - based on the relative abundance of microbial genera in the sample. I demonstrate that microbial abundance can be used as a weak signal to be picked up by machine learning models for classifying colorectal cancer subtypes and note that the genera *Porphyromonas*, *Bilophila*, and *Solobacterium* consistently have high influence on model predictions using SHAP.

Introduction

Burden and Relevance

In the United States, over 2 million cases of cancer are projected to be diagnosed in 2025 and over 600,000 deaths are projected to occur as a result of cancer with the leading cause being lung cancer and the second leading cause being colorectal cancer [1]. Colorectal cancer consists of colon cancer, rectal cancer, and cancers onsetting from both sites combined, and the risk of colorectal cancer increases with age [2]. Although colorectal cancer has historically been more common with older adults, early onset incidence has increased in recent years which motivates the need for wider awareness and earlier detection [3]. Early detection is crucial in preventing advanced onset of colorectal cancer. Colorectal cancer is detected through screening (colonoscopy) and via screening, mortality can be reduced significantly, particularly with screening in younger individuals [4].

Need for Differentiation

In colorectal cancer, it's important to differentiate between colon and rectum cancers. Rectum and colon cancers are treated differently. While primary treatment for both may be surgical removal [5], rectum cancer treatment is supplemented by radiotherapy or chemoradiotherapy prior to surgery while colon cancer is occasionally supplemented by adjuvant chemotherapy [6, 7]. By not differentiating rectum and colon cancers correctly, it can lead to under- or over-treatment such as unnecessary or omittance of radiotherapy or chemoradiotherapy prior to surgery [7]. Outcomes of patients with colon and rectum cancers also differ. Short term mortality is worse for colon cancer patients than for rectal cancer patients, and rectal cancer patients experience more complications following surgery [5]. Given treatment and outcome differences, misdiagnoses have serious implications for the patient.

Machine Learning and Microbiome Data

A strong classification tool is machine learning, and many studies have demonstrated its efficacy in being applied to cancer, specifically in prognosis and prediction [8]. Machine learning refers to the automated detection of meaningful patterns in data [9]. There are many data modalities that can support machine learning, each capturing different aspects of the underlying information and providing unique classification signals. In clinical oncology, imaging and molecular data are

common and can be obtained from liquid or solid tumor biopsies [10]. Another modality to be explored is microbiome data, specifically the gut microbiome. The gut microbiome is the collection of bacteria, viruses, and other microbes that live in the human gut [11]. It is connected to cancer development but the exact link is not yet clear [12]. Specifically, associations have been found between gut microbiota and colorectal cancer risk [13, 14].

Freitas et al. (2023) applied machine learning to microbiome data specifically targeting classification abilities of a Random Forest model in separating the five cancer types present in The Cancer Microbiome Atlas (TCMA) dataset. Their model showed promise in classifying head and neck, stomach, and colon cancer cases but had difficulty with esophageal and rectum cancers [15]. Difficulty arises as the esophagus is anatomically adjacent to the stomach and the rectum is anatomically adjacent to the colon, which causes environmental overlap. These results are consistent with a statistical analysis of the TCMA dataset performed by Wang et al. (2021). The authors found that microbial profiles of upper and lower gastrointestinal (GI) were distinguishable but noted similarities within each group (upper and lower) such that the profile of stomach adenocarcinoma (STAD) was close to esophageal adenocarcinoma (ESCA) and that samples from colon adenocarcinoma (COAD) and rectum adenocarcinoma (READ) nearly overlapped [16]. The similarity between the microbial environments of COAD and READ samples provides a unique challenge in differentiating the two cancers.

Research Objective

Given the importance of differentiating between colon and rectum cancers, the connection between the gut microbiome and colorectal cancer, and the previously noted difficulty in distinguishing colon and rectum adenocarcinoma from the TCMA dataset, the aim of this work is to build off of Freitas et al. (2023) and apply various machine learning methods to the colon and rectum cohorts of the TCMA dataset in isolation with the goal of finding a classifier to adequately separate the two cancer types. A classifier would be adequate if it has high classification scores (such as AUC or accuracy) and reliable predictions, making it potentially useful in a clinical setting.

Data: Source & Analysis

Data Source

My data source is The Cancer Microbiome Atlas (TCMA), a publicly accessible dataset out of Duke University connecting microbiome profiles to samples of five different cancer types: Head-Neck Squamous Cell Carcinoma (HNSC), Esophageal Carcinoma (ESCA), Stomach Adenocarcinoma (STAD), Colon Adenocarcinoma (COAD), and Rectum Adenocarcinoma (READ) [17]. TCMA contains information regarding the prevalence of microbial species across samples from The Cancer Genome Atlas (TCGA), a large-scale publicly-funded cancer genomics project [18]. TCMA can be downloaded by taxonomic level, and I chose to download by genus as this is the most fine-grain of the levels offered and is the taxonomic level of choice of previous work done by Freitas et al. (2023).

Downloading by genus produced two text files:

- sample_metadata.txt, and
- bacteria.sample.relabund.genus.txt

Using the Pandas Library I converted both to CSV and then transposed and merged by sample ID to produce a single CSV out of the two data files. I then filtered the dataset to only include the primary tumor sample type. This produced the following distribution: 155 samples of HNSC, 127 samples of STAD, 125 samples of COAD, 60 samples of ESCA, and 45 samples of READ. Since my objective is to compare just COAD and READ, I removed rows for ESCA, HNSC, and STAD from the dataset as well as any columns that had empty values or zero relative abundance. Microbial abundance is already normalized in the TCMA dataset so no further processing was necessary.

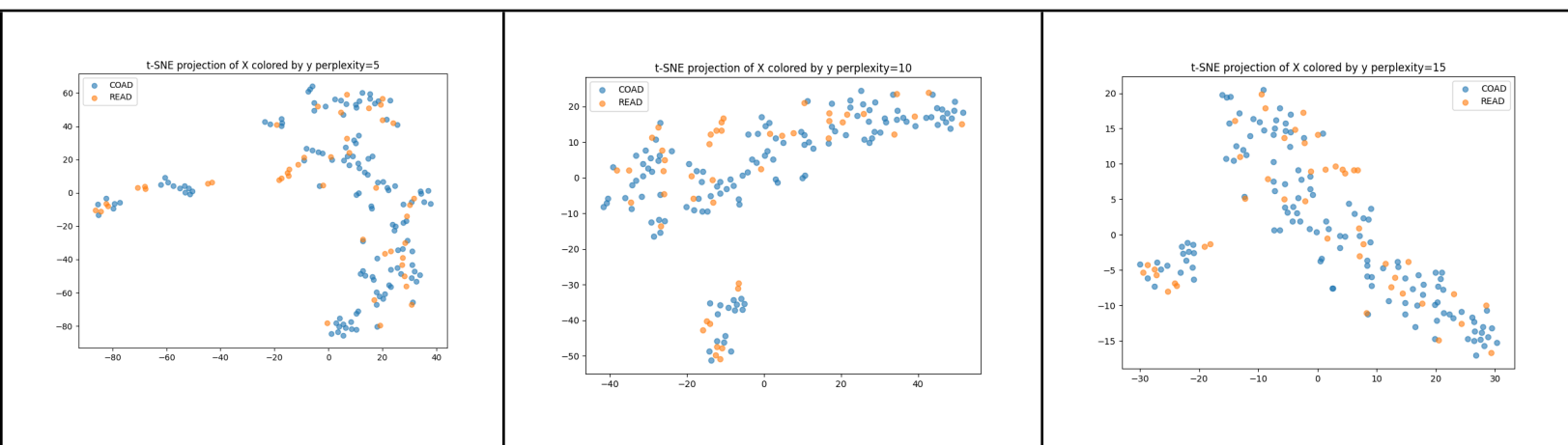
My machine learning task is to use microbial abundance percentages to correctly classify the associated cancer type. In line with this objective, my predictive features are the various microbial genera and their relative abundance and my target variable is the cancer type (COAD or READ). The shape of my features is (170, 92) and my target variable shape is a one-dimension array of length 170 where the dimension of 170 data points represents 125 COAD

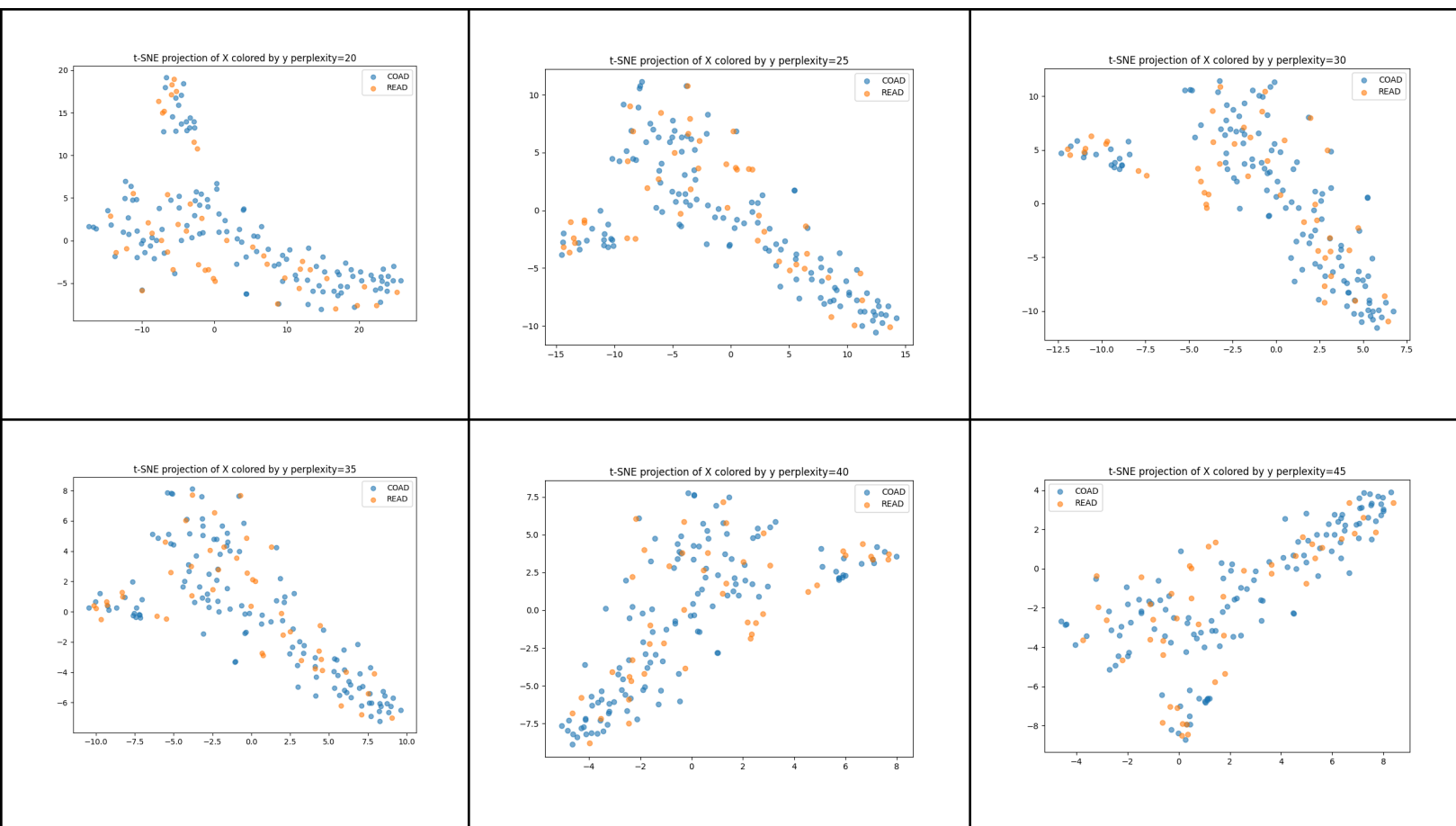
and 45 READ samples and the dimension of 92 is the number of microbial genera whose relevant abundance is represented in the sample.

Exploratory Data Analysis - tSNE

To visualize the dataset to assess potential for separability I applied the t-SNE (t-Distributed Stochastic Neighbor Embedding) algorithm using Scikit-learn. Scikit-learn is a Python library for predictive data analysis. It has many tools available for data analysis and machine learning including classification, regression, clustering, dimensionality reduction, model selection, and preprocessing. The purpose of performing t-SNE is to visualize high dimensional data in low dimensional space, which is particularly useful for cancer datasets that are high dimensional such as ours. t-SNE builds off SNE (Stochastic Neighbor Embedding). It addresses limitations of SNE by using a symmetric cost function and a Student-t distribution to compute similarities, which improves optimization and reduces the crowding problem [19].

t-SNE has a parameter to tune called perplexity which comes from SNE. The t-SNE paper by Maaten and Hinton (2008) states that perplexity can be interpreted as a smooth measure of the effective number of neighbors and suggest a range of 5 to 50 [19]. Using our microbiome features split by their corresponding cancer type, I visualized perplexities 5 through 45 in increments of 5. Per the plots below, there is no obvious separability and often points between classes appear to overlap. This provides an indication that the features for COAD and READ are very similar, or at least difficult to differentiate visually when plotted in low dimensional space. Regardless of this, we'll still attempt to build machine learning classifiers that can determine the separation criteria that is otherwise hard to see with the human eye.





Methods

Overview

To differentiate between COAD and READ I utilized three machine learning models from Scikit-learn. I choose Scikit-learn for its noted simplicity and consistency throughout its machine learning models [20]. The models that I've chosen are Logistic Regression, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). The choice of these models is motivated by an increasing level of complexity. Logistic Regression is a linear model that maps y as a sigmoid function of x with one of its benefits being its simplicity [21]. SVM can be either a linear or non-linear model, specifically in Scikit-learn by specifying either a linear or radial basis function kernel. Generally, SVM works by finding the hyperplanes that separate classes by the largest distance [22]. An MLP is a non-linear model and is the most flexible of the three. The MLP is a simple form of an artificial neural network. A benefit of the MLP is its theoretical ability to represent arbitrary continuous functions when non-linear activations are used. The

universal approximation theorem suggests an ability to approximate any continuous function with arbitrary accuracy using only a single hidden layer paired with non-linear activation [23]. This potentially makes the MLP a promising model.

I split the data using Scikit-learn's train test split function, allocating 85% of the dataset to training and 15% for the holdout set. I set the stratify parameter to ensure that the class proportion is consistent between the training set and the holdout set. These models require various hyperparameters. To optimize these, I used Scikit-learn's GridSearchCV with 5 stratified folds over the training set. This performs cross validation on every combination of specified hyperparameters to find the best set optimizing on a specified score function. The score function I used across models is 'roc_auc', or Area Under the Receiver Operating Characteristic Curve (AUC). Bradley (1997) compared AUC with overall accuracy across six machine learning algorithms on medical diagnostic datasets and recommended AUC due to its favorable properties, including decision threshold independence [24]. Given the class imbalance in our dataset, AUC is a more appropriate and reliable measure for model evaluation.

With this best set of hyperparameters, I perform bootstrapping over 5-fold cross-validation on the training set over 1000 iterations to assess robustness, with each iteration with new random resampling and fold determination. We then assess our model on the holdout set. For both the bootstrapped cross-validation and holdout set results I provide visualization of AUC, noting where the holdout set performance falls within the 95% confidence interval distribution of bootstrapped cross-validation performance as well as a plot of true positive rate (TPR) vs. false positive rate (FPR). I also provide confusion matrix heatmaps to further depict the distribution of predictions made for cross-validation (on average) and the holdout set. The value add of the confusion matrix in addition to AUC to visualize how the model is favoring between the two classes.

On the holdout set performance I provide SHAP (SHapley Additive exPlanations) for feature importance and calculate the Brier score and plot the Calibration Curve for uncertainty analysis. SHAP gives us an importance value for each feature in the model's predictions. It does this by attributing to each feature a value that represents the change in the expected model prediction

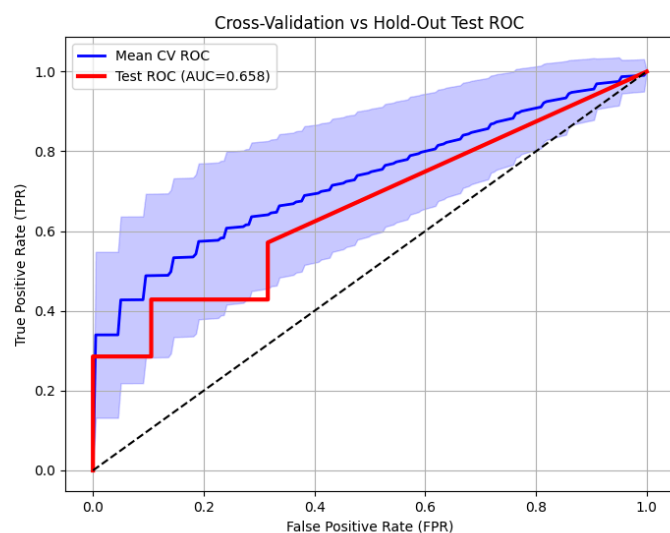
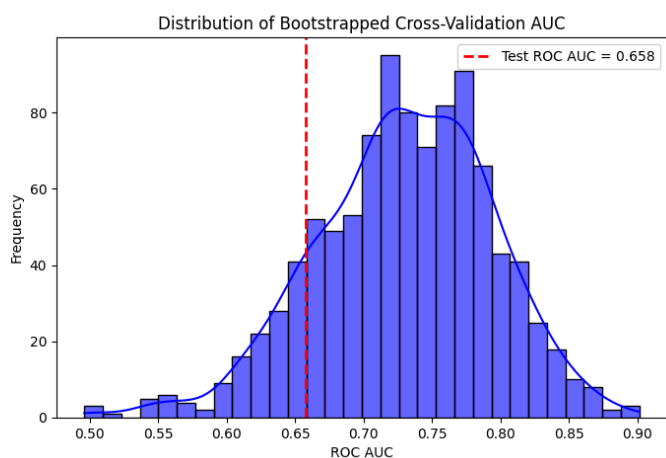
when conditioning on that feature [25]. Brier score is a method for evaluating the accuracy of a set of probabilistic predictions and calculated via Mean Squared Error between the probabilities of the outcomes (between 0 or 1) and the outcomes (0 or 1), where a value closer to zero is desirable such that zero represents perfect accuracy [26, 27]. The Calibration Curve visually depicts the correctness of prediction probabilities. Dawid (1982) describes well-calibrated to mean that assigned probabilities turn out to be the actual probabilities [28]. By evaluating calibration we get a notion of reliability.

Logistic Regression

Hyperparameter Selection via Grid Search (selected parameters highlighted in green)

C	0.001	0.01	0.1	1	10
Penalty	l1	l2	elasticnet		
Solver	saga	lbfgs	liblinear	newton-cg	
L1 Ratio	0.1	0.3	0.5	0.7	0.9
Max Iterations	10	25	50	100	
Class Weight	balanced				

AUC Plots

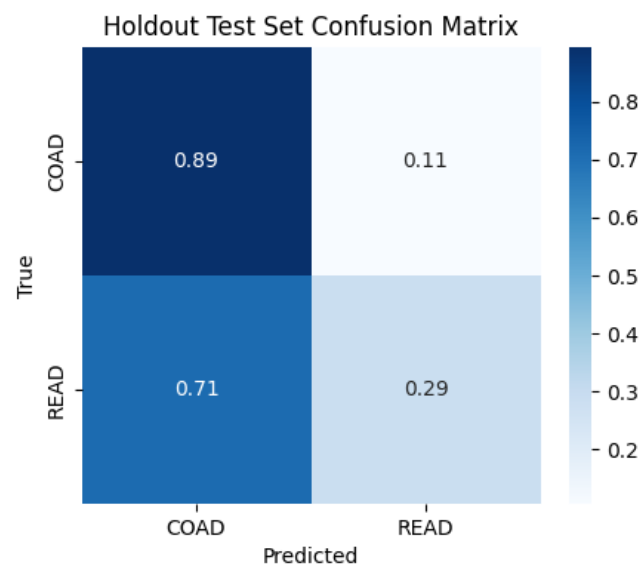
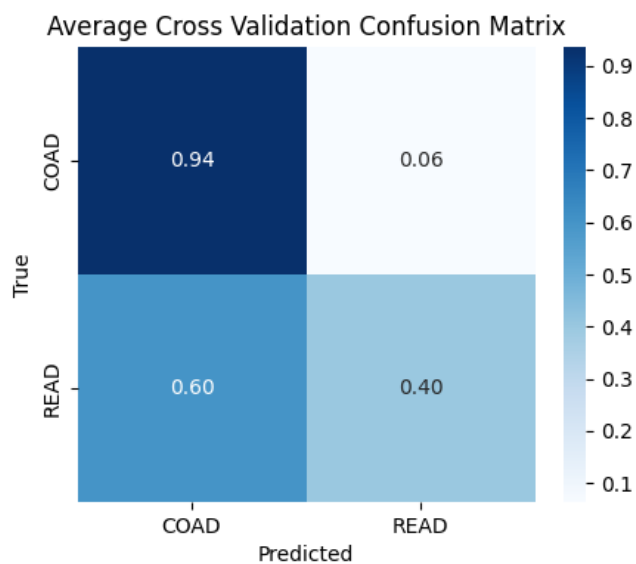


Mean bootstrapped CV ROC AUC: 0.7295

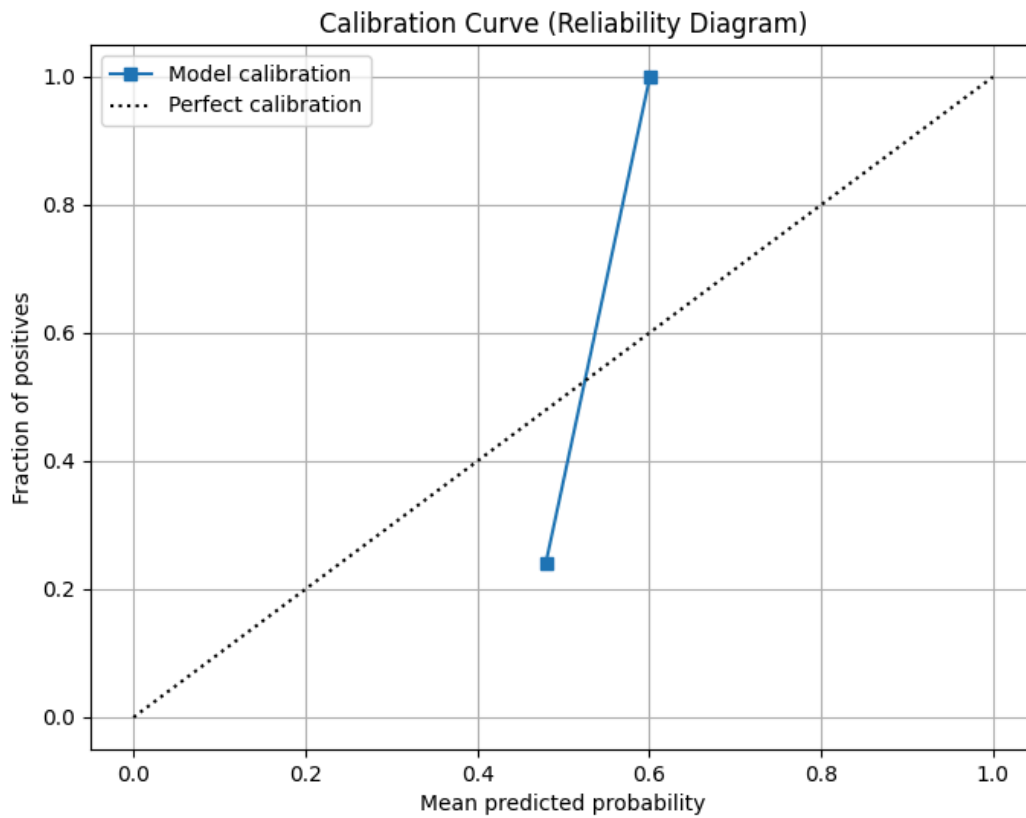
95% confidence interval: [0.6003, 0.8453]

Test set ROC AUC: 0.6579

Confusion Matrix Heatmaps

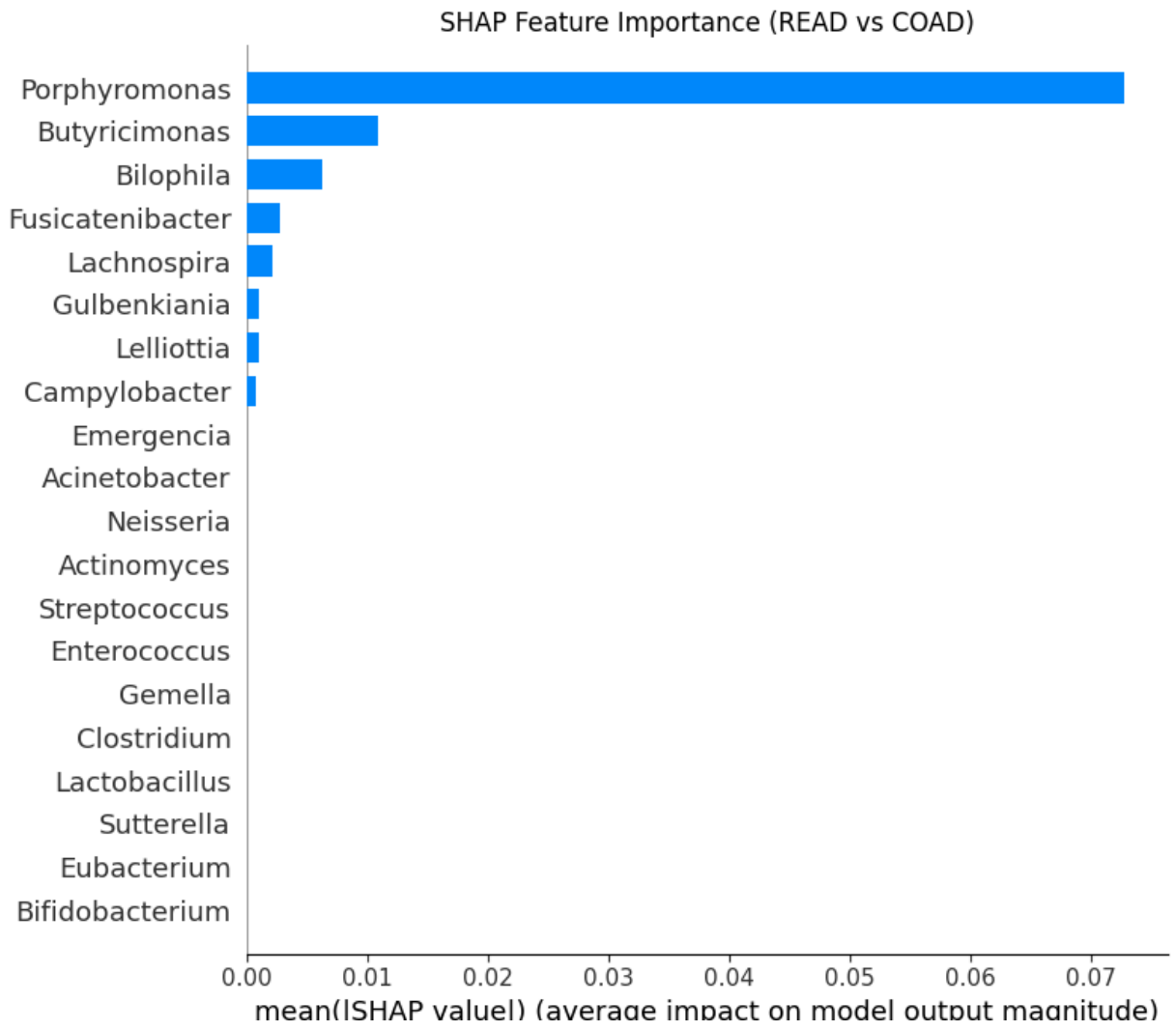


Uncertainty on the Holdout Set - Calibration and Brier



Brier score: 0.2340

SHAP Feature Importance

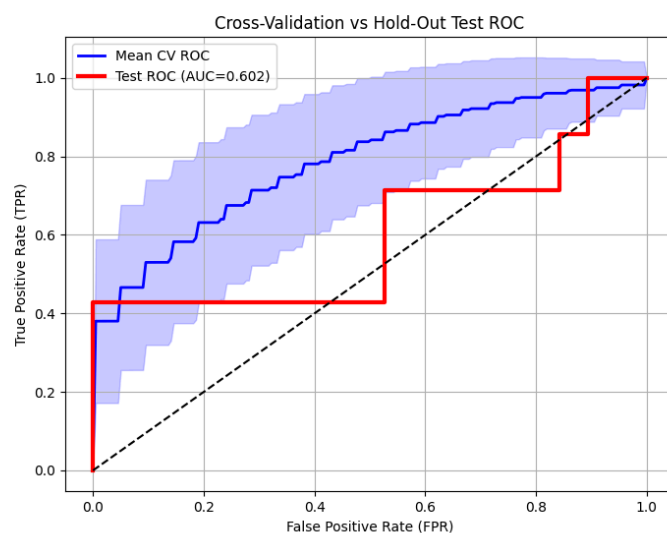
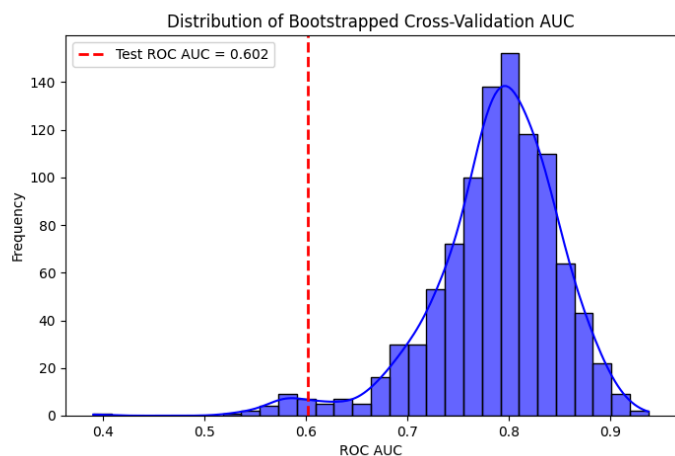


SVM

Hyperparameter Selection via Grid Search (selected parameters highlighted in green)

C	0.01	0.1	1	10	25	50
Gamma	0.00001	0.0001	0.001	scale	auto	
Kernel	rbf	linear	poly	sigmoid		

AUC Plots

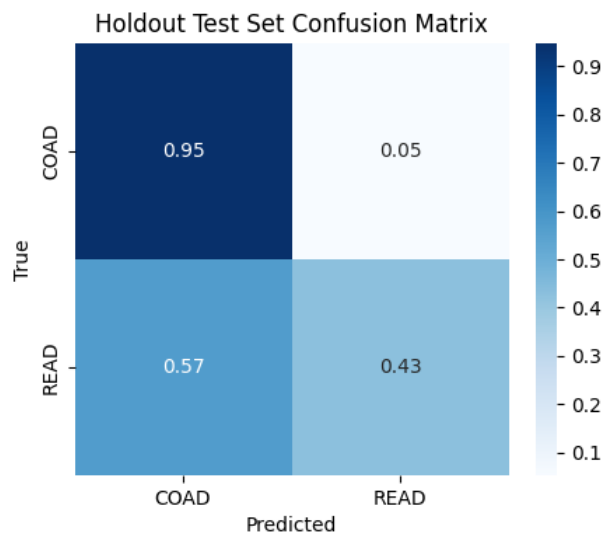
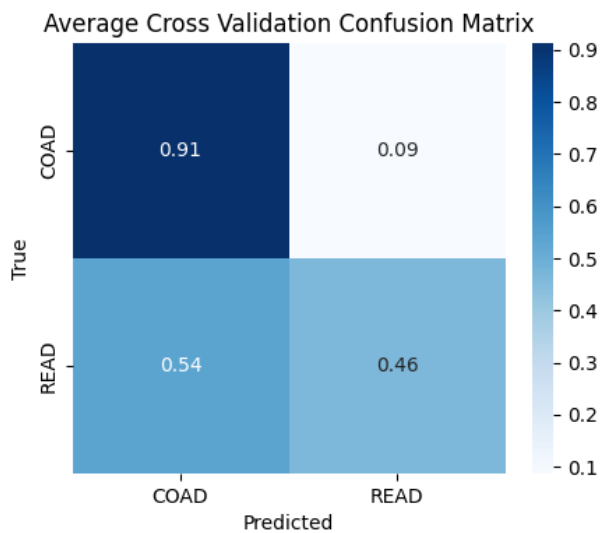


Mean bootstrapped CV ROC AUC: 0.7882

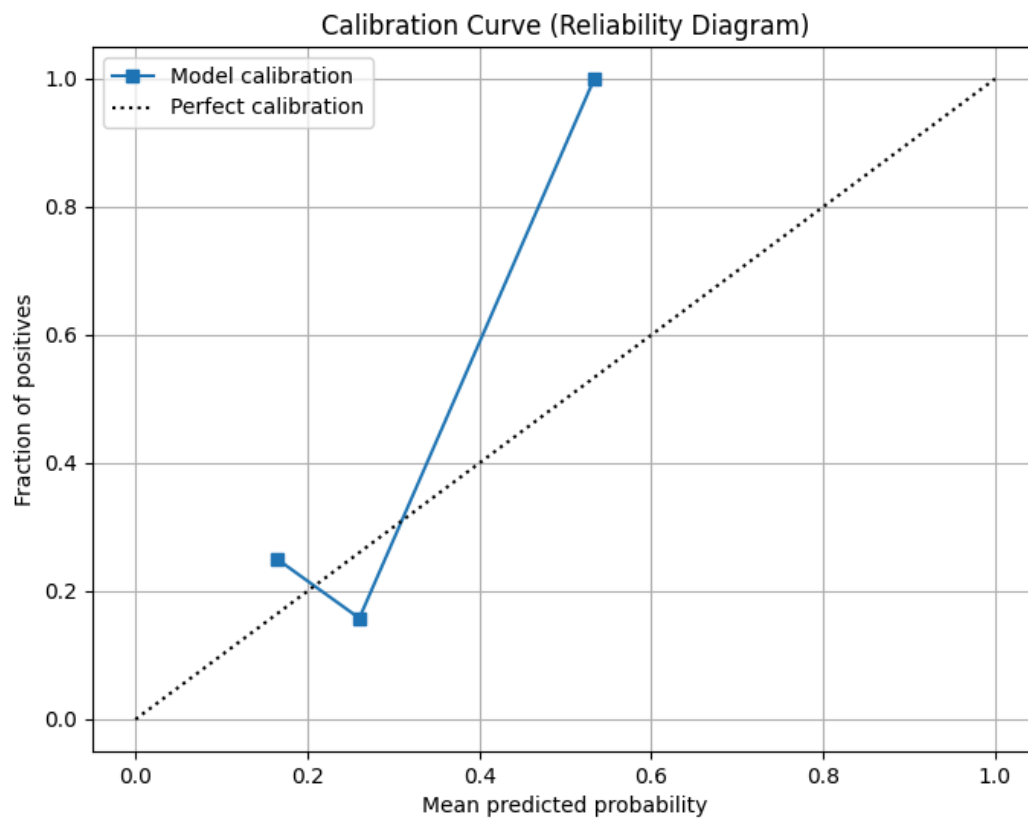
95% confidence interval: [0.6284, 0.8881]

Test set ROC AUC: 0.6015

Confusion Matrix Heatmaps

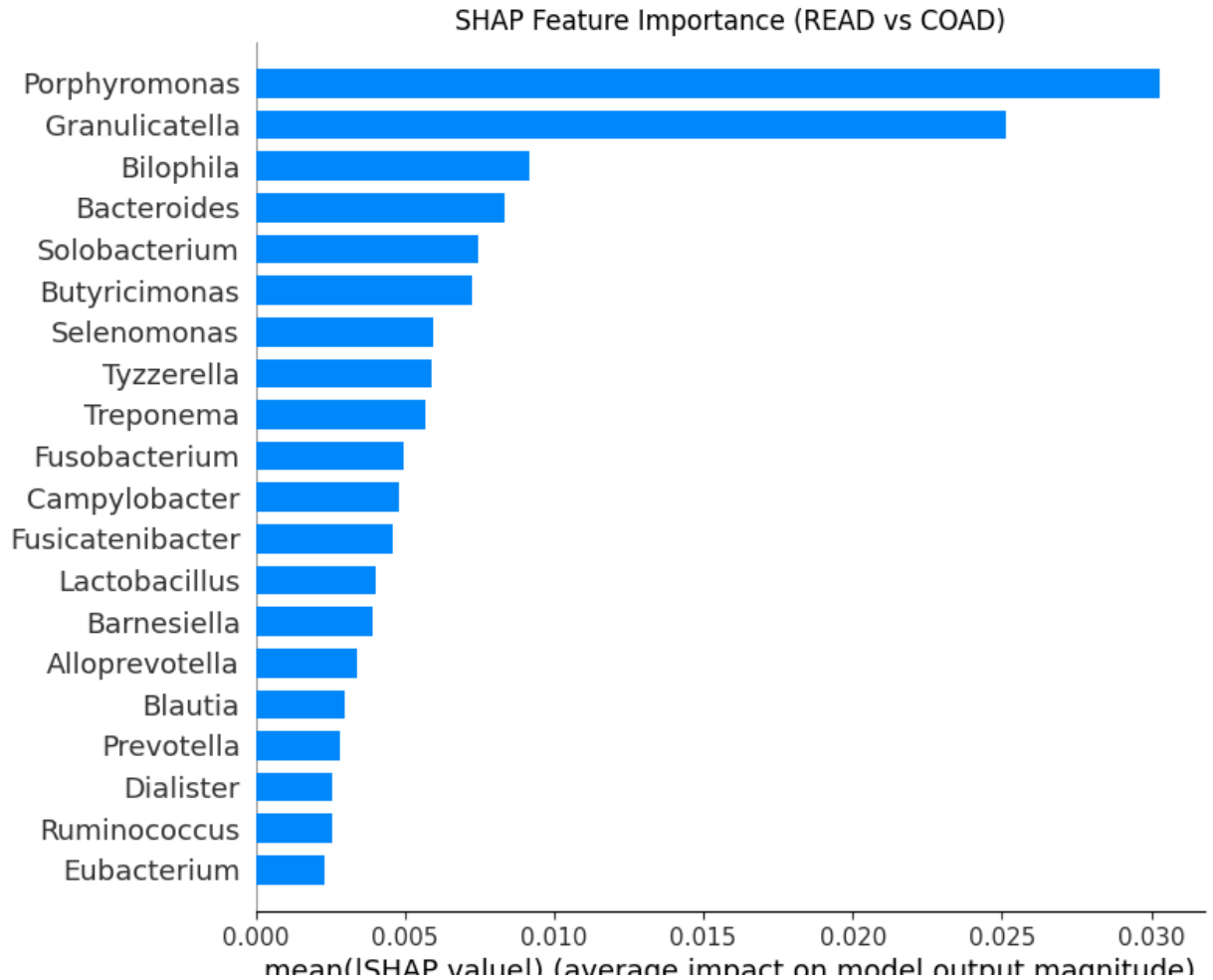


Uncertainty on the Holdout Set - Calibration and Brier



Brier score: 0.1674

SHAP Feature Importance



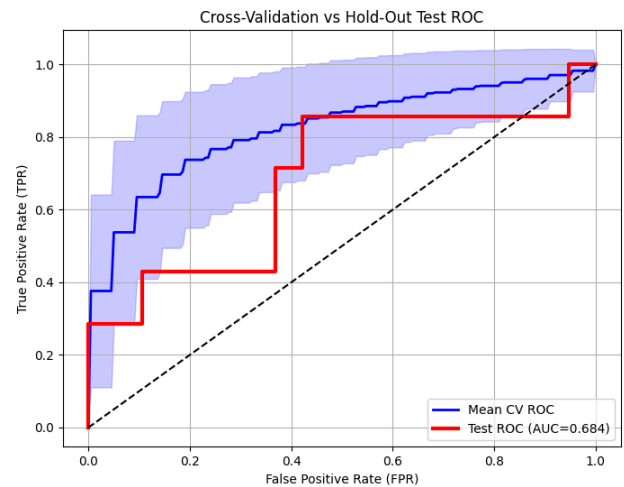
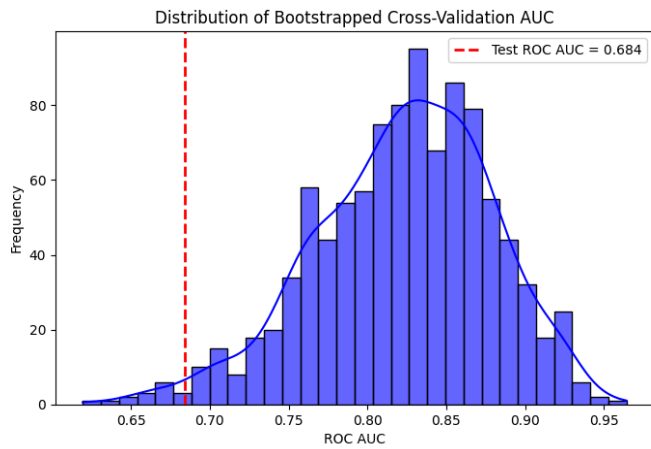
Multi-layer Perceptron (MLP)

Hyperparameter Selection via Grid Search (selected parameters highlighted in green)

Hidden Layer Sizes	(2,)	(5,)	(10,)	(5,2)	(10,2)
Alpha	0.1	1.0	5.0	10	
Initial Learning Rate	0.0001	0.0005	0.001		

Activation	relu	tanh	logistic		
Solver	lbfgs	adam	sgd		
Max Iterations	25	50	75	100	
Early Stopping	True				

AUC Plots

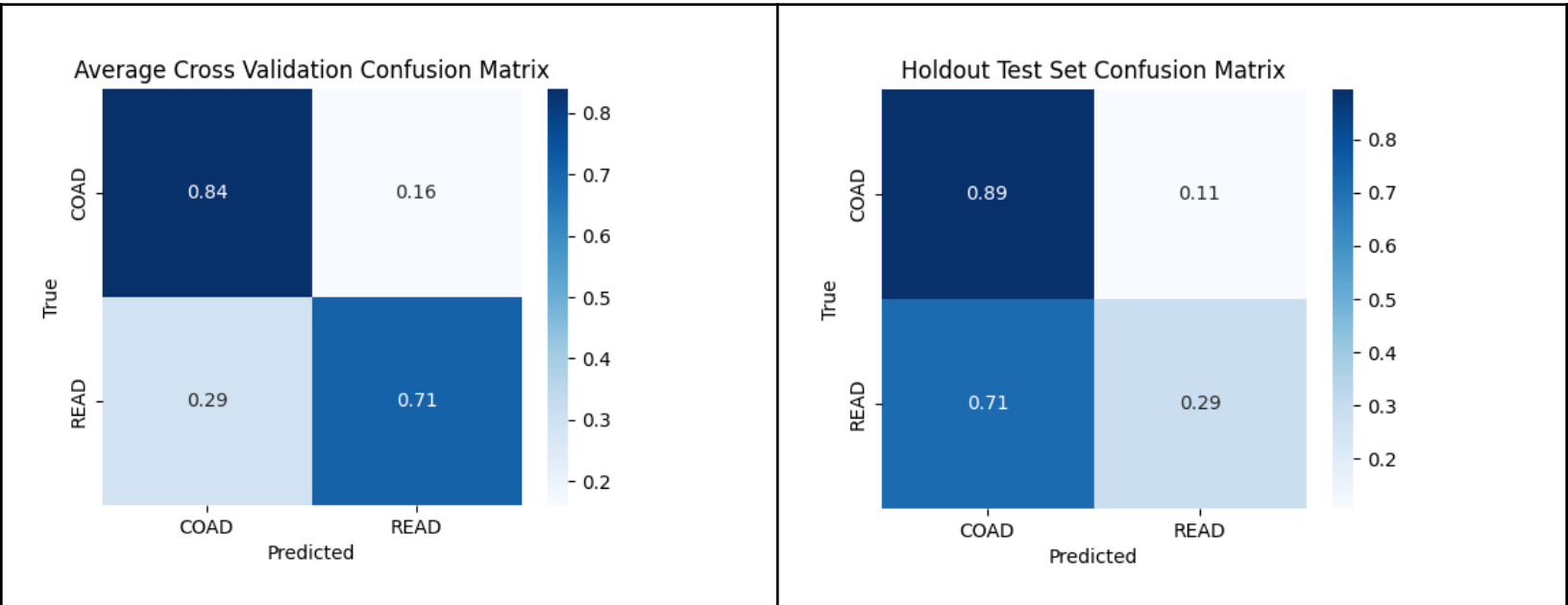


Mean bootstrapped CV ROC AUC: 0.8207

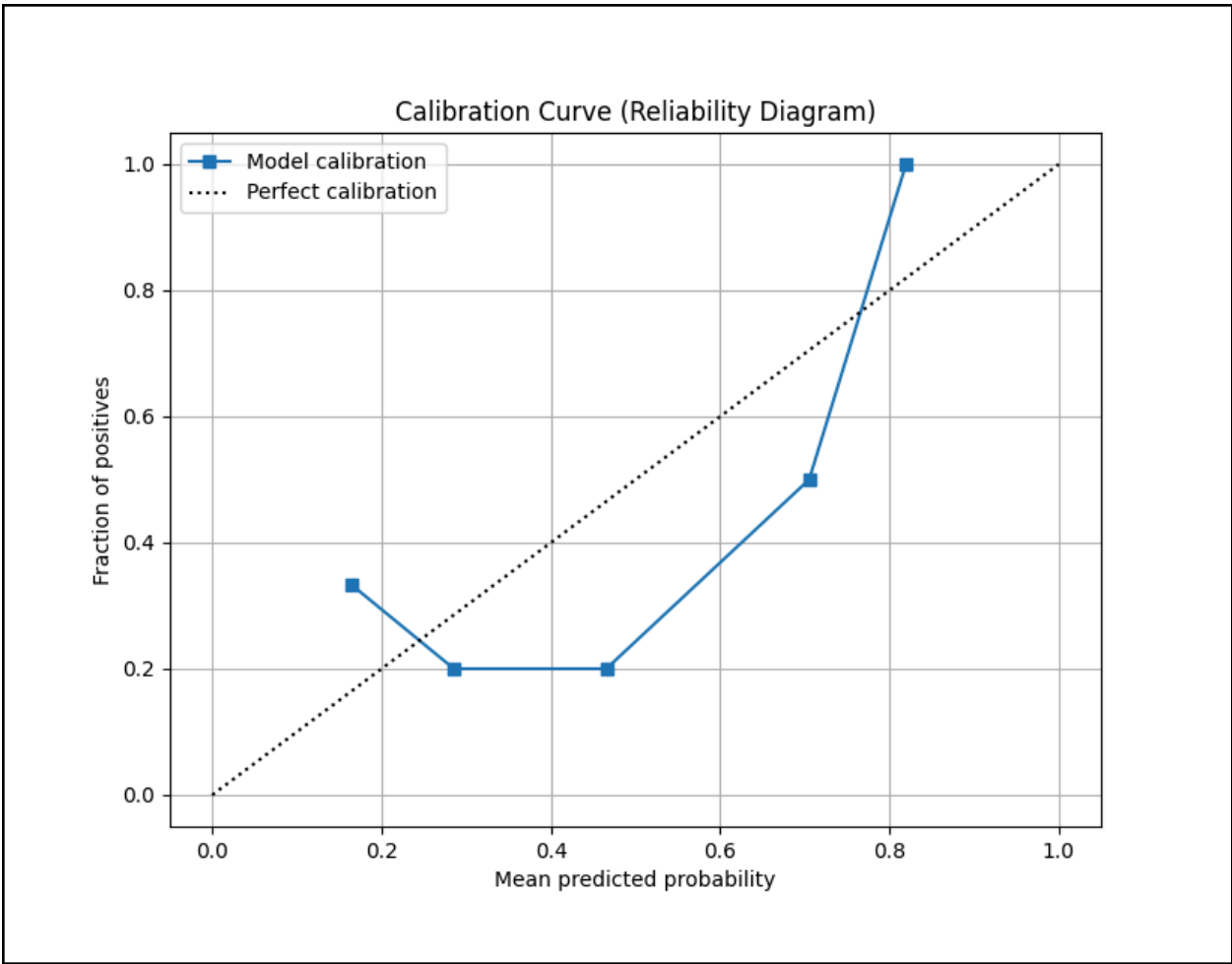
95% confidence interval: [0.6805, 0.9218]

Test set ROC AUC: 0.6842

Confusion Matrix Heatmaps

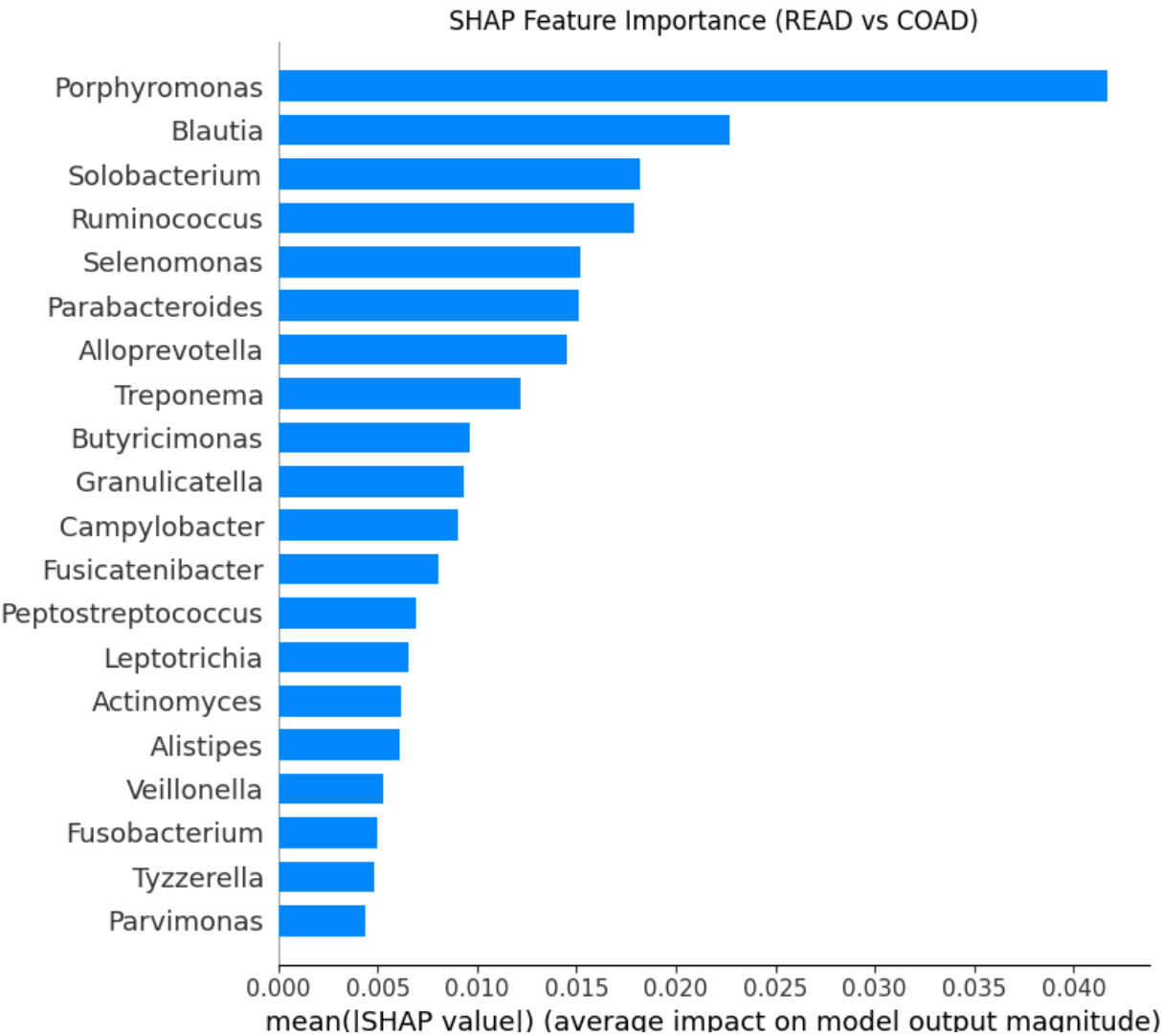


Uncertainty on the Holdout Set - Calibration and Brier



Brier score: 0.1843

SHAP Feature Importance



Results

The below table reports the AUC scores across the three models assessed, indicating cross-validation (mean and 95% confidence interval) and holdout set AUC.

	Mean CV AUC	95% CI CV AUC	Holdout AUC
Logistic Regression	0.7295	[0.6003, 0.8453]	0.6579
SVM	0.7882	[0.6284, 0.8881]	0.6015
MLP	0.8207	[0.6805, 0.9218]	0.6842

MLP performed the best of the three models in terms of AUC for inference during cross-validation and on the holdout set. Following this, SVM performed slightly better in cross-validation than Logistic Regression but Logistic Regression performed better on the holdout set. No single model was statistically significant in outperforming the others at the 95% confidence interval. As visible in the table, each 95% confidence interval is overlapping with the others.

The below table summarizes the recall scores for each class for each model.

	CV COAD, READ Recall	Holdout COAD, READ Recall
Logistic Regression	0.94, 0.4	0.89, 0.29
SVM	0.91, 0.46	0.95, 0.43
MLP	0.84, 0.71	0.89, 0.29

Average cross validation recall across the three models is 0.8967 for COAD and 0.5233 for READ. This means that on average in cross-validation we correctly classify 89.67% of COAD samples as COAD but incorrectly classify 10.33% of COAD samples as READ, and we correctly classify 52.33% of READ samples as READ but incorrectly classify 47.67% of READ samples as COAD. READ recall is skewed by stronger performance from the MLP. In the holdout set

recall is similar to cross-validation for COAD but consistently worse for READ. Average holdout recall across the three models is 0.91 for COAD and 0.3366 for READ. This means that on average in the holdout set we correctly classify 91% of COAD samples as COAD but incorrectly classify the remaining 9% of COAD samples as READ, and we correctly classify 33.66% of READ samples as READ but incorrectly classify the remaining 66.34% of READ samples as COAD.

The below table summarizes uncertainty of the three models:

	Brier Score
Logistic Regression	0.2340
SVM	0.1674
MLP	0.1843

SVM had the best Brier score followed by MLP and then Logistic Regression. As a whole, these are far from perfect. Logistic Regression would nearly be a “fence-sitter”, as randomly assigning 0.5 probably would result in a Brier score of 0.25 [27]. SVM and MLP do better than randomly assigning probabilities of 0.5 but have a lot of room for improvement. The calibration curve plots noted in the Methods section above reflect this sentiment as they don’t fit the diagonal well but are at least in the right general direction.

The below table summarizes the SHAP feature importance analysis giving the top five most important genera for each model, their mean absolute SHAP value, and which cancer the model is influenced to choose on average given higher abundance of the specific genus.

Logistic Regression			
Rank	Genus	Mean Absolute SHAP	Higher Abundance → COAD or READ
1	Porphyromonas	0.0728	COAD

2	Butyricimonas	0.0109	READ
3	Bilophila	0.0063	COAD
4	Fusicatenibacter	0.0028	COAD
5	Lachnospira	0.0022	COAD
SVM			
Rank	Genus	Mean Absolute SHAP	Higher Abundance → COAD or READ
1	Porphyromonas	0.0303	COAD
2	Granulicatella	0.0251	READ
3	Bilophila	0.0091	COAD
4	Bacteroides	0.0083	COAD
5	Solobacterium	0.0075	READ
MLP			
Rank	Genus	Mean Absolute SHAP	Higher Abundance → COAD or READ
1	Porphyromonas	0.0417	COAD
2	Blautia	0.0227	COAD
3	Solobacterium	0.0182	READ
4	Ruminococcus	0.0179	READ
5	Selenomonas	0.0152	COAD

Of the genera, Porphyromonas had the highest mean absolute SHAP score in all three models. Higher abundance of this genus led the model to predict COAD. Bilophila appeared in the top

five at rank three for both the Logistic Regression and SVM model but was not among the top five for MLP. On average, higher abundance of *Bilophila* caused the model to predict COAD in both cases. *Solobacterium* appeared in the top five at rank five and three for the SVM and MLP models, respectively, but was not in the top five for Logistic Regression. On average, higher abundance of *Solobacterium* caused the model to predict READ in both cases. Other genera amongst the top five lists did not appear in the top five for the other models.

Discussion

Using the TCMA dataset, microbial abundance can be used as a weak signal to be picked up by machine learning models for classifying colorectal cancer into the subtypes COAD and READ. Comparison of cross-validation and holdout set performance of models (Logistic Regression, SVM, and MLP) consistently showed overfitting, indicating difficulty in generalization regardless of model complexity. The models did perform better than random, but a large contributing factor is high recall in COAD. Recall for READ was consistently poor on the holdout set. Moreover, our models had low certainty in their estimates making even correct predictions unreliable, as indicated by the Brier scores. This is an undesirable result given our previously noted differences in treatments and patient outcomes of these cancer types, so I would not recommend clinical use in isolation. The models should aim for at least 0.80 AUC for clinical usefulness, but my implementations of Logistic Regression, SVM, and MLP all scored less than 0.7 on the holdout set which would be regarded as not clinically adequate [29].

The mediocre performance of the three machine learning models is not surprising given the dataset size and relative class imbalance. Our total number of samples is 170 with 73.53% COAD and 26.47% READ (125 and 45 samples, respectively), which is a large difference and introduces inherent difficulty. Considering the train/test split further reducing what's available for training, this does not leave many examples to learn how to correctly predict READ, which was a downfall in my models (as we saw poor recall for READ). A possible area for future work to improve prediction performance is to apply threshold optimization via a technique called Generalized Threshold Shifting Procedure (GHOST). GHOST aims to maximize a statistical measure called Cohen's kappa on the training set and with the corresponding threshold use it to predict the test set. The authors of the GHOST technique found it to be beneficial for most

classifiers analyzed [30]. Instead of adjusting the threshold, another potential direction to improve performance is to apply Synthetic Minority Over-sampling Technique (SMOTE), which does a special type of over-sample of the minority class (which in our case would be READ) to create synthetic examples and under-samples the majority class (for the TCMA dataset, COAD). The authors of this technique found it can improve performance [31].

Our feature importance analysis revealed *Porphyromonas*, *Bilophila*, and *Solobacterium* as consistently ranking high. Kerdreux et al. (2023) notes a possible link between colorectal cancer and *Porphyromonas* finding significantly different levels of this genus in faecal samples from colorectal cancer patients compared to their control [32]. My findings further their research by suggesting that the link would be more associated with colon cancer than with rectum cancer. My findings provide similar additions to other existing work supporting associations in colorectal cancer with *Bilophila* [33] and *Solobacterium* [34].

Lastly, this work covers COAD and READ from the TCMA dataset in isolation but there is yet to be work to distinguish between STAD and ESCA in isolation. This is just as interesting of a challenge given that these two cancers are also anatomically adjacent like COAD and READ but instead of lower gastrointestinal cancers, STAD and ESCA are upper gastrointestinal cancers. While machine learning models applied to these upper gastrointestinal cancers may achieve similar AUC as in distinguishing COAD and READ (as the same limitations apply: the large class imbalance within a small dataset), it would be interesting to see the distribution of genera deemed important by SHAP and if it is consistent across models. The models should be still evaluated for clinical relevance and the most important genera determined by SHAP should be compared to any existing papers discussing their relevance to upper gastrointestinal cancers.

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